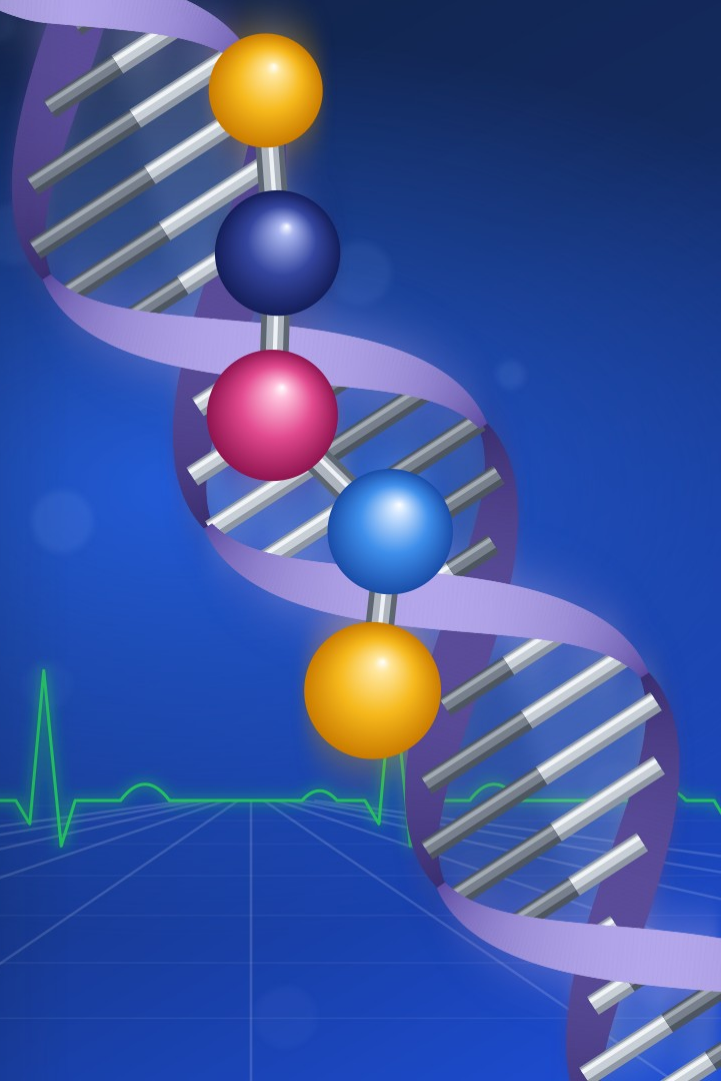
DHRUV
DIAGNOSTICS

Athlete and Physique Health Check-Up

Cardiac, muscle, hormone, longevity and tendon assessment for athletes and bodybuilders.



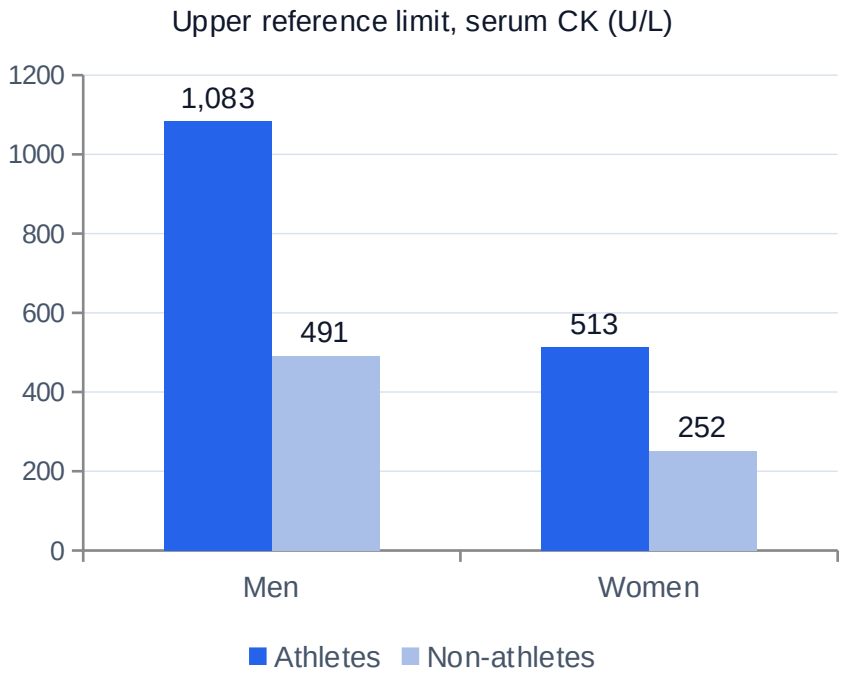
NABL accredited since 2011 | Technical overview for clinicians and coaches

Athletes run higher CK than non-athletes

Mougios (2007) measured serum creatine kinase in 483 male and 245 female athletes. The upper reference limit in athletes was about twice that of non-athletes, and footballers ran higher still.

A CK of 900 U/L, flagged as high on a standard report, can be ordinary for a male athlete in season.

The same problem affects haemoglobin, creatinine, AST, ALT and ECG voltage. Reports in this plan are read against the athlete's sport and training phase.



Source: Mougios V. Br J Sports Med 2007;41:674-678. 97.5th percentile, 37 °C.

What each plan includes

Investigation	Base	Performance	Elite
Resting ECG, blood pressure, anthropometry	●	●	●
Haemogram with reticulocytes, ferritin, vitamin D, B12	●	●	●
Glucose, HbA1c, lipid profile, liver, kidney with cystatin C, TSH	●	●	●
Muscle panel: CK, CK-MB, LDH, myoglobin	CK only	●	●
Apo B, Lp(a), hs-CRP, homocysteine, fasting insulin		●	●
NT-proBNP		●	●
Hormone panel: testosterone, LH, FSH, SHBG, oestradiol, prolactin		●	●
hs-troponin, cortisol, IGF-1, urine albumin:creatinine			●
Treadmill stress test (TMT)			●
Tendon and ligament ultrasound by an MSK radiologist		1 region	Up to 3 regions
Urine drugs-of-abuse screen	Add-on	Add-on	Add-on
Physician review of the full report	●	●	●

Treadmill stress test, Bruce protocol

Stage	Speed (km/h)	Grade (%)	METs at end, approx.
1	2.7	10	5
2	4.0	12	7
3	5.5	14	10
4	6.8	16	13
5	8.0	18	15
6	8.8	20	17
7	9.7	22	19

Each stage lasts 3 minutes. METs are estimates and vary by equation.

Reported

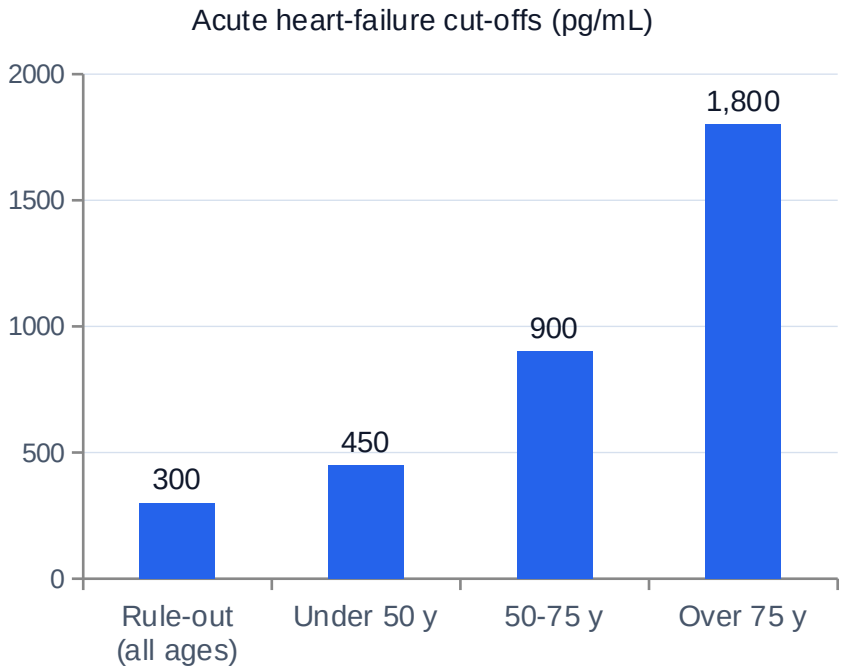
- Exercise time, stage reached and estimated METs
- Peak heart rate as % of age-predicted maximum
- Blood-pressure response at each stage
- ST-segment change and arrhythmia
- Heart-rate recovery at 1 minute (12 bpm or less is abnormal)

Advised for athletes over 35, with a family history of early heart disease, or with symptoms on exertion. An equivocal test is referred for echocardiography or cardiology review.

NT-proBNP as a personal cardiac baseline

What it measures
A fragment released when heart muscle is stretched by pressure or volume load.

Use in athletes
Marathons and long rides raise it for a day or more. We sample at rest, at least 24 hours after hard training, record the value as a baseline and compare it at each repeat visit.



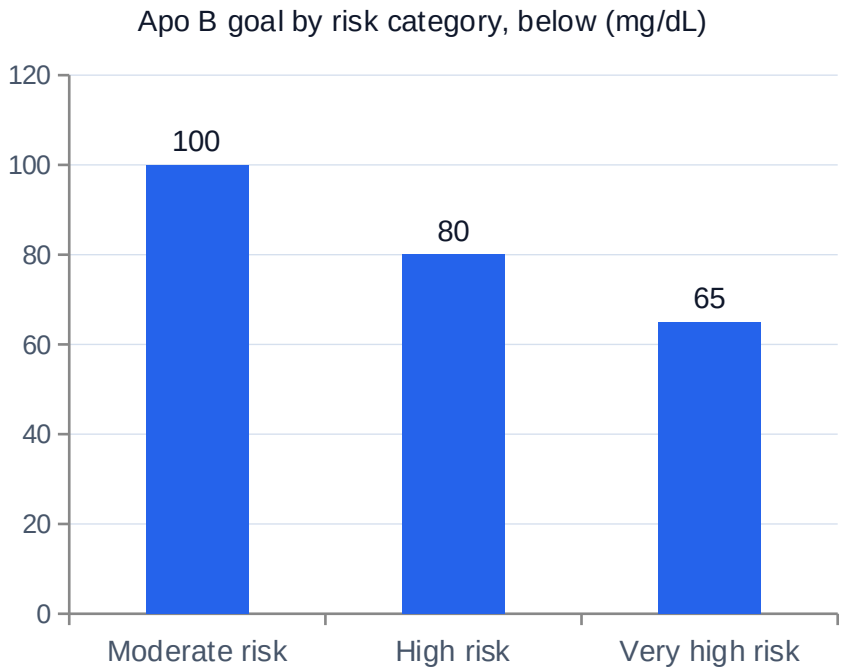
Cut-offs from ESC Heart Failure Association guidance, derived in breathless patients. For reference only.

Apo B counts atherogenic particles

Each LDL, VLDL, IDL and Lp(a) particle carries one Apo B molecule, so Apo B is a direct particle count.

LDL cholesterol can look normal while Apo B is high. We see this with high triglycerides, insulin resistance, very low-carbohydrate diets and anabolic steroid use.

Lp(a) is largely inherited. It is measured once in adult life and is included with the first Apo B test.



Goals from ESC/EAS 2019 dyslipidaemia guidelines. The physician assigns the risk category.

Muscle damage and recovery markers

Marker	What it reflects	How we read it in athletes
CK (CPK), total	Leak from damaged muscle fibres; peaks 24 to 72 h after eccentric or unfamiliar work	Judged against athlete ranges. Above 5 times the upper limit with weakness or dark urine needs same-day review
CK-MB	Cardiac isoform, also present in regenerating skeletal muscle	A raised CK-MB with very high total CK is usually skeletal; troponin settles the question
hs-troponin I or T	Specific to heart muscle	Can rise briefly after endurance events. A rise at rest goes to cardiology
Myoglobin, serum and urine	Early marker, rises within hours of muscle injury	Myoglobin in urine with high CK signals risk to the kidneys
LDH	Muscle, red cells, liver	Supports CK. Foot-strike haemolysis in runners also raises it
AST, ALT	Found in muscle as well as liver	CK is checked before a raised AST or ALT is called liver injury
Creatinine with cystatin C	Creatinine tracks muscle mass and creatine intake	Cystatin C gives a truer kidney function in muscular people
Potassium, phosphate, uric acid	Released when muscle breaks down	Checked when CK is markedly raised

Hormones and blood in strength athletes

Marker	Why it is measured
Haemoglobin, haematocrit, reticulocytes	Androgens can push haematocrit up and thicken the blood. Endurance athletes may run low iron stores
Testosterone (total, free), LH, FSH	Shows suppression of the body's own production and its recovery after a cycle stops
SHBG, oestradiol, prolactin	Needed to interpret testosterone; oestradiol and prolactin explain many side effects
Cortisol (8 am), testosterone:cortisol	Read with training load as a guide to under-recovery
IGF-1	Growth-hormone axis; also raised by high protein intake
TSH, free T4	Thyroid function affects weight, energy and heart rate
GGT, bilirubin, ALT	Oral steroids and some supplements stress the liver
Oestradiol, LH, FSH, ferritin in women	Low energy availability (RED-S) disturbs menstrual hormones and bone health

Results are confidential. Telling us about supplements or enhancers helps the physician interpret them correctly.

Markers linked to long-term health

Marker	What it adds
Fitness in METs, from the TMT	Lower fitness carried about 5 times the mortality of elite fitness in 122,007 treadmill patients
Apo B and Lp(a)	Lifetime artery risk; Lp(a) is genetic and measured once
hs-CRP	Low-grade inflammation. Repeated if raised by recent hard training or infection
HbA1c, fasting glucose, fasting insulin	HOMA-IR shows insulin resistance years before diabetes
Homocysteine	Linked to vascular risk; often reflects low B12 or folate
Uric acid	Associated with high blood pressure, gout and metabolic syndrome
Cystatin C eGFR, urine albumin:creatinine	Early kidney and small-vessel damage
Vitamin D, B12, ferritin	Deficiencies are common in India and affect bone, nerves and energy

Fitness data: Mandsager K et al. JAMA Netw Open 2018;1(6):e183605. Hazard ratio 5.04, low vs elite fitness.

Ultrasound of tendons and ligaments

Region	Structures examined
Shoulder	Rotator cuff, long head of biceps, subacromial bursa
Elbow	Common extensor and flexor origins, distal biceps
Hip and groin	Gluteal tendons, adductor origin, iliopsoas
Knee	Patellar and quadriceps tendons, MCL, LCL
Ankle and foot	Achilles, ATFL, peroneal tendons, plantar fascia

Findings reported

- Tendon thickness compared with the other side
- Hypoechoic areas and loss of fibre pattern
- New vessels on Doppler (neovascularity)
- Partial tears and calcific deposits
- Bursal fluid
- Ligament gapping on stress views

Performed and reported by a radiologist trained in musculoskeletal ultrasound. High-frequency linear probe, no radiation. The first scan is the baseline for return-to-play decisions.

How official doping control works

When banned	WADA Prohibited List class, 2026
At all times	S0 non-approved substances, S1 anabolic agents, S2 peptide hormones and growth factors (EPO, hGH), S3 beta-2 agonists, S4 hormone and metabolic modulators, S5 diuretics and masking agents, M1 blood manipulation, M2 chemical and physical manipulation, M3 gene and cell doping
In competition	S6 stimulants, S7 narcotics (tramadol since 2024), S8 cannabinoids (CBD excepted), S9 glucocorticoids
In some sports	P1 beta-blockers, for example archery, shooting and golf

Who tests and how

- NADA India tests Indian athletes; the International Testing Agency runs testing at the Olympic Games
- Urine (at least 90 mL) and blood, split into A and B samples
- Analysed only at WADA-accredited laboratories; in India, NDTL New Delhi
- Athlete Biological Passport tracks blood, steroid and growth-hormone markers over time
- Olympic samples may be stored and re-tested for up to 10 years

Dhruv Diagnostics does not carry out doping control and cannot certify an athlete as clean.

Urine drug screen and WADA status

Drug group screened	WADA class	Banned	Note
Amphetamine, methamphetamine	S6 stimulants	In competition	
MDMA (ecstasy)	S6; substance of abuse	In competition	
Cocaine	S6; substance of abuse	In competition	Screen detects benzoylecgonine
Cannabinoids (THC)	S8; THC is a substance of abuse	In competition	CBD alone is allowed; CBD products can contain THC
Opiates (morphine)	S7 narcotics	In competition	Screen also reacts to codeine, which is not banned
Tramadol	S7 narcotics	In competition	Banned since 1 January 2024
Methadone, buprenorphine	S7 narcotics	In competition	
Benzodiazepines, barbiturates	Not on the List	No	Relevant to health and safety only

Immunoassay is a screen. A positive result needs confirmation by GC-MS or LC-MS/MS. This screen is for clinical and personal use and is not a doping test.

Recovery Club for women members

Open to women members only at present. Men's access has not opened yet.

Sauna

Dry heat after training. Opened to a member once her blood pressure and ECG or TMT have been reviewed.

Spa

Massage and hydrotherapy between sessions to ease muscle soreness and stiffness.

Health club

Supervised strength, conditioning and mobility work planned around the check-up findings.

Sauna not advised with	Precautions for every member
Pregnancy, uncontrolled high blood pressure, recent heart event, fever or acute illness	Water and electrolytes before and after; no alcohol; leave at any dizziness

Preparing for the visit

Instruction	Timing	Protects the result of
Overnight fast, water allowed	10 to 12 h	Glucose, insulin, lipids
No hard, eccentric or very long training	48 to 72 h before	CK, myoglobin, AST, ALT, troponin, NT-proBNP
Blood drawn early	8 to 10 am	Testosterone, cortisol
Stop high-dose biotin (hair and skin supplements)	72 h before	Thyroid, hormone and troponin immunoassays
Tell us about creatine use	At booking	Creatinine (cystatin C is added)
Bring a list of medicines, supplements and any TUE	On the day	Hormone panel, drug screen, physician review
Light breakfast after the blood draw; wear running shoes	TMT at least 2 h after eating	Treadmill stress test

Repeat every 6 to 12 months so results can be compared with the athlete's own baseline.

Sources

1. World Anti-Doping Agency. 2026 Prohibited List, in force 1 January 2026.
2. World Anti-Doping Agency. Athlete Biological Passport Operating Guidelines.
3. Mougios V. Reference intervals for serum creatine kinase in athletes. Br J Sports Med 2007;41:674-678.
4. Mandsager K et al. Cardiorespiratory fitness and long-term mortality among adults undergoing exercise treadmill testing. JAMA Netw Open 2018;1(6):e183605.
5. Mach F et al. 2019 ESC/EAS Guidelines for the management of dyslipidaemias. Eur Heart J 2020;41:111-188.
6. Mueller C et al. Heart Failure Association of the ESC practical guidance on the use of natriuretic peptide concentrations. Eur J Heart Fail 2019;21:715-731.
7. Kronenberg F et al. Lipoprotein(a) in atherosclerotic cardiovascular disease: EAS consensus statement. Eur Heart J 2022;43:3925-3946.
8. Cole CR et al. Heart-rate recovery immediately after exercise as a predictor of mortality. N Engl J Med 1999;341:1351-1357.
9. Press Information Bureau, Government of India. NDTL regains WADA accreditation, December 2021.

Laboratory tests are performed in a NABL-accredited laboratory. The check-up is a screening programme and does not replace a consultation with your own doctor. Dhruv Diagnostics does not perform doping control; official testing is carried out by NADA India at WADA-accredited laboratories.

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Ramdaspeth map